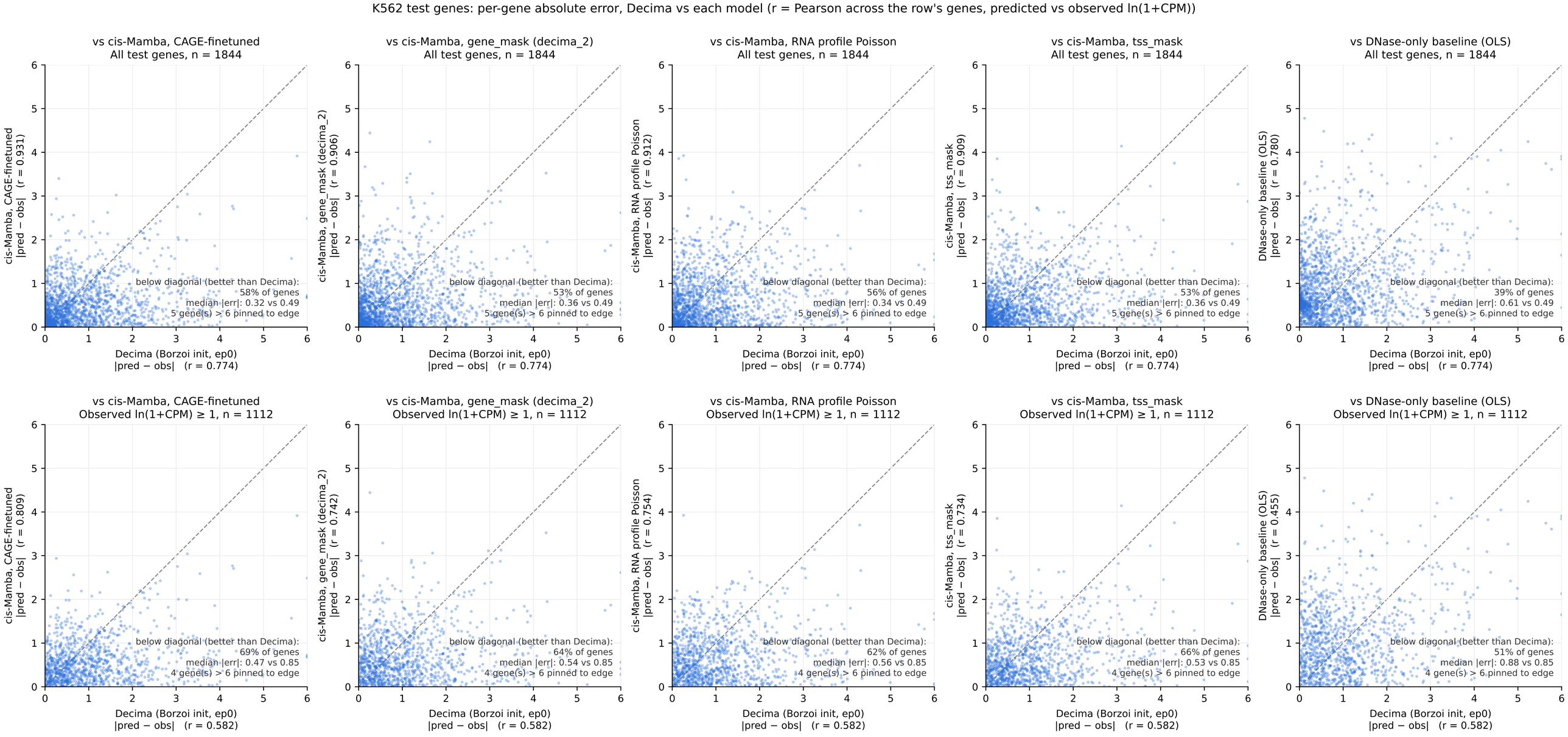

K562 test genes: per-gene absolute error, Decima vs each model (r = Pearson across the row's genes, predicted vs observed ln(1+CPM))
vs cis-Mamba, CAGE-finetuned
All test genes, n = 1844
cis-Mamba, CAGE-finetuned |pred − obs| (r = 0.931)
below diagonal (better than Decima): 58% of genes
median |err|: 0.32 vs 0.49
5 gene(s) > 6 pinned to edge
vs cis-Mamba, gene_mask (decima_2)
All test genes, n = 1844
cis-Mamba, gene_mask (decima_2) |pred − obs| (r = 0.906)
below diagonal (better than Decima): 53% of genes
median |err|: 0.36 vs 0.49
5 gene(s) > 6 pinned to edge
vs cis-Mamba, RNA profile Poisson
All test genes, n = 1844
cis-Mamba, RNA profile Poisson |pred − obs| (r = 0.912)
below diagonal (better than Decima): 56% of genes
median |err|: 0.34 vs 0.49
5 gene(s) > 6 pinned to edge
vs cis-Mamba, tss_mask
All test genes, n = 1844
cis-Mamba, tss_mask |pred − obs| (r = 0.909)
below diagonal (better than Decima): 53% of genes
median |err|: 0.36 vs 0.49
5 gene(s) > 6 pinned to edge
vs DNase-only baseline (OLS)
All test genes, n = 1844
DNase-only baseline (OLS) |pred − obs| (r = 0.780)
below diagonal (better than Decima): 39% of genes
median |err|: 0.61 vs 0.49
5 gene(s) > 6 pinned to edge
Decima (Borzoi init, ep0) |pred − obs| (r = 0.774)
vs cis-Mamba, CAGE-finetuned
Observed ln(1+CPM) ≥ 1, n = 1112
cis-Mamba, CAGE-finetuned |pred − obs| (r = 0.809)
below diagonal (better than Decima): 69% of genes
median |err|: 0.47 vs 0.85
4 gene(s) > 6 pinned to edge
vs cis-Mamba, gene_mask (decima_2)
Observed ln(1+CPM) ≥ 1, n = 1112
cis-Mamba, gene_mask (decima_2) |pred − obs| (r = 0.742)
below diagonal (better than Decima): 64% of genes
median |err|: 0.54 vs 0.85
4 gene(s) > 6 pinned to edge
vs cis-Mamba, RNA profile Poisson
Observed ln(1+CPM) ≥ 1, n = 1112
cis-Mamba, RNA profile Poisson |pred − obs| (r = 0.754)
below diagonal (better than Decima): 62% of genes
median |err|: 0.56 vs 0.85
4 gene(s) > 6 pinned to edge
vs cis-Mamba, tss_mask
Observed ln(1+CPM) ≥ 1, n = 1112
cis-Mamba, tss_mask |pred − obs| (r = 0.734)
below diagonal (better than Decima): 66% of genes
median |err|: 0.53 vs 0.85
4 gene(s) > 6 pinned to edge
vs DNase-only baseline (OLS)
Observed ln(1+CPM) ≥ 1, n = 1112
DNase-only baseline (OLS) |pred − obs| (r = 0.455)
below diagonal (better than Decima): 51% of genes
median |err|: 0.88 vs 0.85
4 gene(s) > 6 pinned to edge
Decima (Borzoi init, ep0) |pred − obs| (r = 0.582)